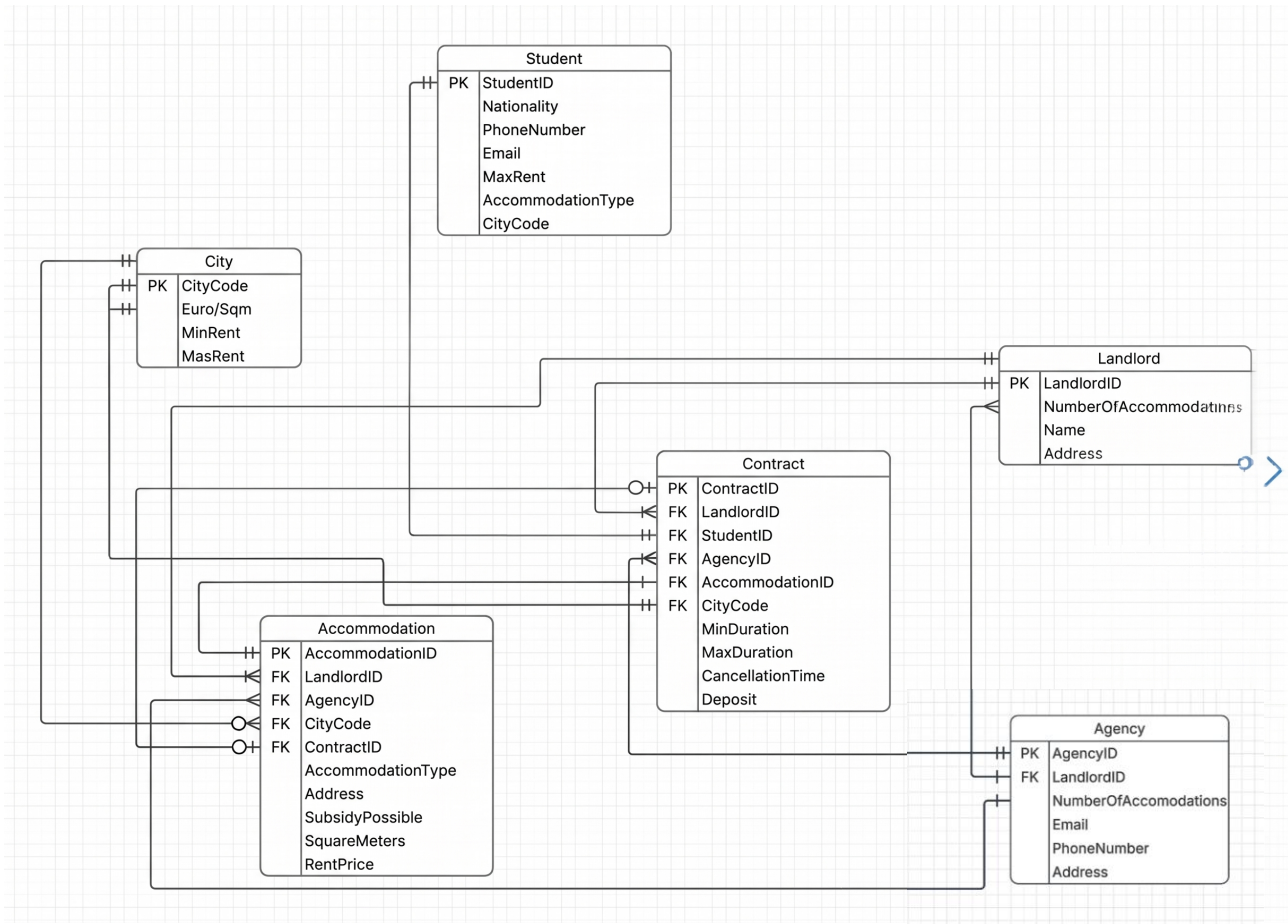


Databases Assignment 2

Normalisation & ERD

Before Normalisation:



Problems:

- Address is stored as one text column in Accommodation, Landlord, and Agency.
- Student.AccommodationType holds only one value, but a student may accept several types.
- Contract carries LandlordID, AgencyID, and CityCode — all already determined by the accommodation.
- Accommodation.ContractID and Contract.AccommodationID point at each other (circular FK).
- Agency.LandlordID implies an agency belongs to one landlord — the wrong direction.
- NumberOfAccommodations is stored in Landlord and Agency, though it can be counted from Accommodation.

Mock Data Tables Before Normalisation:

(data generated by AI)

CITY

CityCode	Euro/Sqm	MinRent	MaxRent
AMS	32,50 €	550,00 €	1400,00 €
UTR	27,00 €	480,00 €	1150,00 €
RTM	24,00 €	450,00 €	1100,00 €
GRO	20,50 €	350,00 €	850,00 €
EHV	22,00 €	420,00 €	950,00 €

STUDENT

StudentID	Nationality	PhoneNumber	Email	MaxRent	AccommodationType	CityCode
1	Dutch	+31 6 2184 5530	d.devries@student.uva.nl	900,00 €	Studio	AMS
2	Dutch	+31 6 4471 0926	s.bakker@students.uu.nl	550,00 €	Room	UTR
3	Dutch	+31 6 3058 7712	l.vandenberg@student.eur.nl	1100,00 €	Apartment	RTM
4	Dutch	+31 6 5129 3348	e.visser@student.rug.nl	400,00 €	Room	GRO
5	Dutch	+31 6 1893 6604	t.mulder@student.tue.nl	700,00 €	Studio	EHV

LANDLORD

LandlordID	NumberOfAccommodations	Name	Address
1	8	Jan de Vries	Keizersgracht 412, 1016 GC Amsterdam
2	3	Sanne Bakker	Oudegracht 118, 3511 AX Utrecht
3	5	Pieter Janssen	Witte de Withstraat 57, 3012 BN Rotterdam
4	2	Femke Smit	Herestraat 94, 9711 LM Groningen
5	4	Bram Hendriks	Stratumseind 21, 5611 ER Eindhoven

AGENCY

AgencyID	LandlordID	NumberOfAccommodations	Email	PhoneNumber	Address
1	1	45	info@grachtenwonen.nl	+31 20 624 1187	Leidsestraat 36, 1017 PB Amsterdam
2	2	28	contact@domstadkamers.nl	+31 30 231 4490	Lange Viestraat 12, 3511 BK Utrecht
3	3	33	verhuur@maashousing.nl	+31 10 412 7735	Coolsingel 88, 3011 AG Rotterdam
4	4	19	info@noorderkamers.nl	+31 50 313 6620	Oude Ebbingestraat 45, 9712 HC Groningen
5	5	22	info@lichtstadwonen.nl	+31 40 244 9051	Vestdijk 70, 5611 CD Eindhoven

ACCOMMODATION

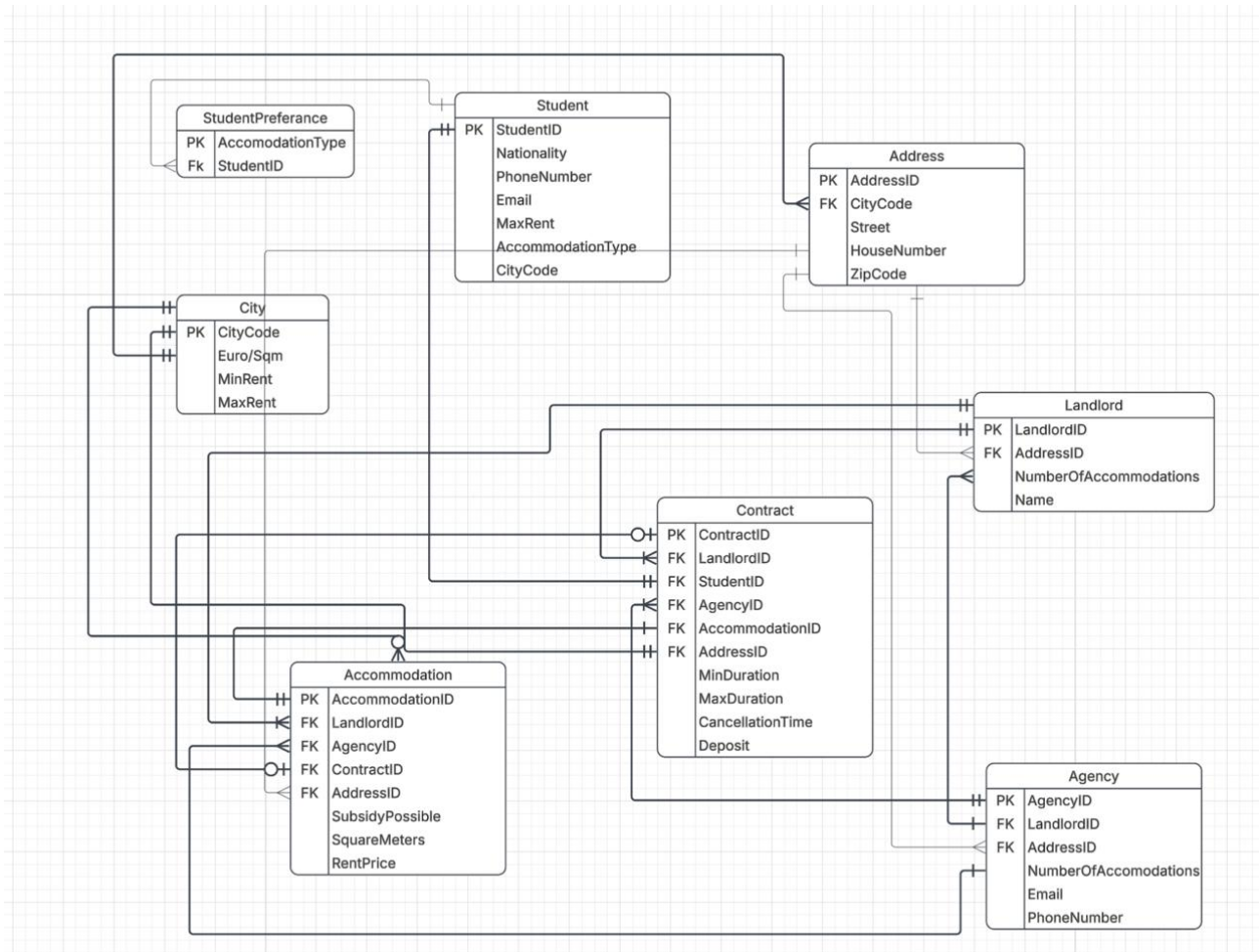
AccommodationID	LandlordID	AgencyID	CityCode	ContractType	Address	SubsidPossible	SquareMeters	RentPrice
1	1	1	AMS	1 Studio	Kinkerstraat 142-2, 1053 ED Amsterdam	Yes	26	845,00 €
2	2	2	UTR	2 Room	Biltstraat 77, 3572 AK Utrecht	No	18	486,00 €
3	3	3	RTM	3 Apartment	Nieuwe Binnenweg 210-B, 3021 GN Rotterdam	No	45	1080,00 €
4	4	4	GRO	4 Room	Nieuwe Ebbingestraat 23, 9712 NC Groningen	No	18	369,00 €
5	5	5	EHV	5 Studio	Kruisstraat 58, 5612 CJ Eindhoven	Yes	30	660,00 €

CONTRACT

ContractID	LandlordID	StudentID	AgencyID	AccommodationID	CityCode	MinDuration	MaxDuration	CancellationTime	Deposit
1	1	1	1	1	AMS	12	24	1	1690,00 €
2	2	2	2	2	UTR	6	12	1	486,00 €
3	3	3	3	3	RTM	12	24	2	2160,00 €
4	4	4	4	4	GRO	6	12	1	369,00 €
5	5	5	5	5	EHV	12	24	1	1320,00 €

After 1NF :

The first form of normalisation is about having only atomic values per cell. In order to ensure this we needed to create an address entity, since a whole address with zip code and street number etc. is a list and not practical in the use of a database (you can see these violations in tables above marked in red). Therefore we created the address entity and linked it with the FK AddressID to the tables that require an address. Also we created a table for student preferences because a student can like several accommodation types like that we don't violate the atomic property of the first normalisation.



Mock Data Tables after 1NF:

(data generated by AI)

Changes are marked yellow.

AgencyID	LandlordID	AddressID	NumberOfAccommodations	Email	PhoneNumber
1	1	11	45	info@grachtenwonen.nl	+31 20 624 1187
2	2	12	28	contact@domstadkamers.nl	+31 30 231 4490
3	3	13	33	verhuur@maashousing.nl	+31 10 412 7735
4	4	14	19	info@noorderkamers.nl	+31 50 313 6620
5	5	15	22	info@lichtstadwonen.nl	+31 40 244 9051

ADDRESS

AddressID	CityCode	Street	HouseNumber	ZipCode
1	AMS	Kinkerstraat	142-2	1053 ED
2	UTR	Biltstraat	77	3572 AK
3	RTM	Nieuwe Binnenweg	210-B	3021 GN
4	GRO	Nieuwe Ebbingestraat	23	9712 NC
5	EHV	Kruisstraat	58	5612 CJ
6	AMS	Keizersgracht	412	1016 GC
7	UTR	Oudegracht	118	3511 AX
8	RTM	Witte de Withstraat	57	3012 BN
9	GRO	Herestraat	94	9711 LM
10	EHV	Stratumseind	21	5611 ER
11	AMS	Leidsestraat	36	1017 PB
12	UTR	Lange Viestraat	12	3511 BK
13	RTM	Coolsingel	88	3011 AG
14	GRO	Oude Ebbingestraat	45	9712 HC
15	EHV	Vestdijk	70	5611 CD

LANDLORD-1

LandlordID	AddressID	NumberOfAccommodations	Name
1	6	8	Jan de Vries
2	7	3	Sanne Bakker
3	8	5	Pieter Janssen
4	9	2	Femke Smit
5	10	4	Bram Hendriks

AccommodationID	LandlordID	AgencyID	ContractID	AddressID	SubsidyPossible	SquareMeters	RentPrice
1	1	1	1	1	Yes	26	845,00 €
2	2	2	2	2	No	18	486,00 €
3	3	3	3	3	No	45	1080,00 €
4	4	4	4	4	No	18	369,00 €
5	5	5	5	5	Yes	30	660,00 €

ContractID	LandlordID	StudentID	AgencyID	AccommodationID	AddressID	MinDuration	MaxDuration	CancellationTime	Deposit
1	1	1	1	1	1	12	24	1	1690,00
2	2	2	2	2	2	6	12	1	486,00 €
3	3	3	3	3	3	12	24	2	2160,00
4	4	4	4	4	4	6	12	1	369,00 €
5	5	5	5	5	5	12	24	1	1320,00

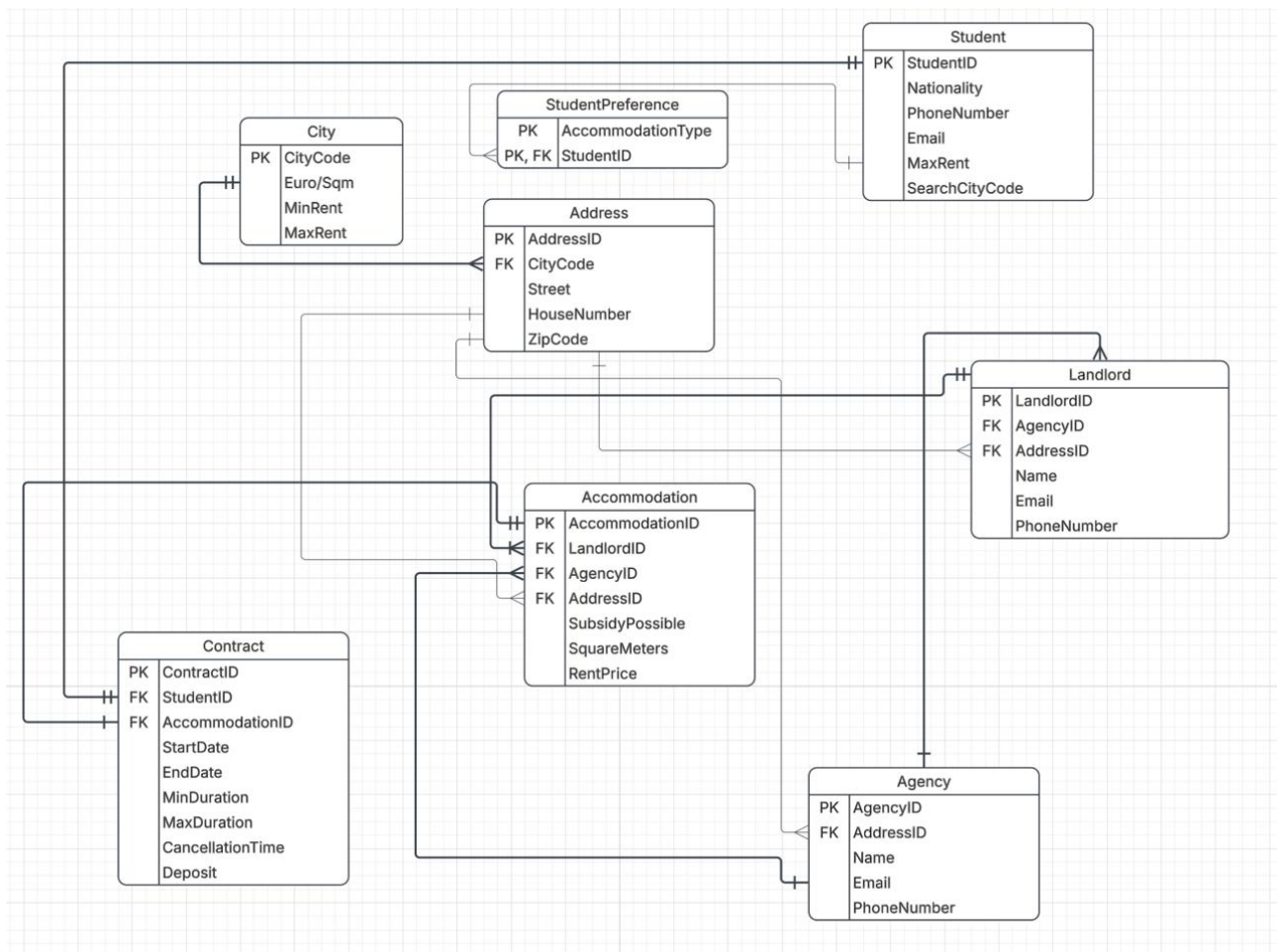
2NF

The second form of normalisation is about composite keys and that they need to be both dependent on the entity that they belong. We choose to not to use composite keys therefore our database didn't require a form 2 normalisation.

3NF:

A table is in the third form of normalisation if its non-key attributes depend on the key attribute and nothing but the key attribute. This is violated by the contract table that contains several IDs (LandlordID, AgencyID, CityCode, AdressID) since all those IDs are dependent on Accommodation we changed them to only Accommodation and removed them from Contract. Furthermore we were able to delete NumberOfAccomodations from Landlord and Agency because that is possible to count via join with Accommodation. Also we removed the landlord from agency and put an agency ID to a Landlord since the dependence is in this direction because the Landlord acquires an Agency. Last we deleted accommodationType because it doubled with StudentPreferences.

After 3NF:



Mock Data Tables after 3NF:

(data generated by AI)

Changes are marked pink.

AccommodationType	StudentID
Studio	1
Room	2
Apartment	3
Room	4
Studio	5

StudentID	Nationality	PhoneNumber	Email	MaxRent	SearchCityCode
1	Dutch	+31 6 2184 5530	d.devries@student.uva.nl	900,00 €	AMS
2	Dutch	+31 6 4471 0926	s.bakker@students.uu.nl	550,00 €	UTR
3	Dutch	+31 6 3058 7712	l.vandenberg@student.eur.nl	1100,00 €	RTM
4	Dutch	+31 6 5129 3348	e.visser@student.rug.nl	400,00 €	GRO
5	Dutch	+31 6 1893 6604	t.mulder@student.tue.nl	700,00 €	EHV

LandlordID	AgencyID	AddressID	Name	Email	PhoneNumber
1	1	6	Jan de Vries	jan.devries@ziggo.nl	+31 6 1047 3382
2	2	7	Sanne Bakker	sanne.bakker@kpnmail.nl	+31 6 2290 6614
3	3	8	Pieter Janssen	p.janssen@xs4all.nl	+31 6 3371 8420
4	4	9	Femke Smit	femke.smit@hotmail.nl	+31 6 4816 2297
5	5	10	Bram Hendriks	bram.hendriks@ziggo.nl	+31 6 5563 0718

AgencyID	AddressID	Name	Email	PhoneNumber
1	11	Grachten Wonen	info@grachtenwonen.nl	+31 20 624 1187
2	12	Domstad Kamers	contact@domstadkamers.nl	+31 30 231 4490
3	13	Maas Housing	verhuur@maashousing.nl	+31 10 412 7735
4	14	Noorder Kamers	info@noorderkamers.nl	+31 50 313 6620
5	15	Lichtstad Wonen	info@lichtstadwonen.nl	+31 40 244 9051

AccommodationID	LandlordID	AgencyID	AddressID	SubsidyPossible	SquareMeters	RentPrice
1	1	1	1	Yes	26	845,00 €
2	2	2	2	No	18	486,00 €
3	3	3	3	No	45	1080,00 €
4	4	4	4	No	18	369,00 €
5	5	5	5	Yes	30	660,00 €

ContractID	StudentID	AccommodationID	StartDate	EndDate	MinDuration	MaxDuration	CancellationTime	Deposit
1	1	1	2025-09-01	2027-08-31	12	24	1	1690,00 €
2	2	2	2026-02-01	2027-01-31	6	12	1	486,00 €
3	3	3	2024-09-01	2026-08-31	12	24	2	2160,00 €
4	4	4	2026-09-01	2027-08-31	6	12	1	369,00 €
5	5	5	2025-02-01	2027-01-31	12	24	1	1320,00 €

4NF:
No violation.